

Operating System

(41TRC2)

IT IV Semester

Submitted by

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Lab Assignment 3

Aim: To create shell scripts for the following questions.

To Perform: To code and solve the following problems.

To Submit: Provide shell scripts for the following:

1. Find the Largest of Three Numbers.

```
echo "Enter three numbers: " read a
b c if [ $a -ge $b ] && [ $a -ge $c then
echo "Largest number is $a" elif [ $b
-ge $a ] && [ $b -ge $c ]; then echo
"Largest number is $b" else echo
"Largest number is $c" fi
```

2. Check if a Year is a Leap Year. #!/bin/bash echo "Enter a year:" read year if (year % 4 0 year % 100 !=0) || (year % 400 0)); then echo "\$year is a leap year." else echo "\$year is not a leap year."

3. Check if Triangle is Valid.

```
#!/bin/bash
echo "Enter three angles of a triangle:"
read a b c sum=$((a + b + c)) if [ $sum -
eq 180 ]; then echo "It is a valid
triangle."
else echo "It is not a valid
triangle." fi
```

4. Check Character Type.

```
#!/bin/bash
echo "Enter a character:" read
char if "$char" [a-zA-Z] I];
then echo "It is an alphabet."
elif "$char" [0-9] then echo
"It is a digit." else echo "It is a
special character."
```

5. Calculate Profit or Loss.

```
#!/bin/bash echo "Enter Cost Price:" read cp
echo "Enter Selling Price:" read sp if [ $sp -gt $cp ]; then echo "Profit:
$((($sp-$cp))" elif [ $cp -gt $sp ]; then echo "Loss: $((($cp-$sp))" else
echo "No Profit, No Loss." fi
```
6. Print Even and Odd Numbers from 1 to 10.

```
#!/bin/bash echo "Even
numbers:" for i in {2..10..2}; do
echo $i; done echo "Odd
numbers:" for i in {1..9..2}; do
echo $i; done
```
7. Print Multiplication Table

```
#!/bin/bash echo "Enter a number:" read
n
for i in {1..10}; do echo
"$n x $i = $((n * i))" done
```
8. Factorial of a Number.

```
#!/bin/bash echo "Enter a number:" read n fact=1
for (( i=1; i<=n; i++ )); do fact=$((fact * i)) done echo "Factorial of $n is
$fact"
```
9. Sum of Even Numbers from 1 to 10.

```
#!/bin/bash
sum=0 for i in {2..10..2}; do sum=$((sum + i))
done echo "Sum of even numbers from 1 to 10 is
$sum"
```
10. Sum of Digits of a Number.

```
#!/bin/bash echo
"Enter a number:" read
num
sum=0 while [ $num -gt 0 ];
do sum=$((sum + num %
10)) num=$((num / 10))
done echo "Sum of digits is
$sum"
```
11. Basic Calculator.

```
#!/bin/bash echo "Enter two numbers:" read a b
```

```

echo "Choose operation: +- * /"
read op case $op in
    "+") echo "Result: $((a + b))" ;;
    "-") echo "Result: $((a - b))" ;;
    "*") echo "Result: $((a * b))" ;;
    "/") echo "Result: $((a / b))" ;;
    * ) echo "Invalid operation" ;;
Esac

```

12. Print Days of the Week

```

#!/bin/bash days=("Sunday" "Monday" "Tuesday" "Wednesday"
"Thursday" "Friday"
"Saturday") for day in "${days[@]"; do
echo $day; done

```

13. Print First 4 Months with 31 Days

```

#!/bin/bash
months=("January" "March" "May" "July")
for month in "${months[@]"; do
echo $month; done

```

14. Using Functions

(a) Check Armstrong Number

```

#!/bin/bash
is_armstrong() { num=$1
sum=0 n=${#num} for ((
i=0; i<n; i++ do
digit=${num:i:1}
sum=$((sum + digit* *n))
done
[[ $sum -eq $num ]] && echo "$num is an Armstrong number" || echo
"$num is not an Armstrong number"

```

```

echo "Enter a number:"
read num is_armstrong

```

```

$num      (b)      Check
Palindrome #!/bin/bash
is_palindrome() {
num=$1      rev=$(echo
$num | rev)
[[ $num -eq $rev ]] && echo "$num is a palindrome" || echo "$num is
not a palindrome"

```

```

echo "Entera number:"
read      num
is_palindrome $num

```

(c) Fibonacci Series

```

#!/bin/bash
fibonacci() { a=0b=1 echo -
n "$a $b" for (( i=2; i<=$1;
i++ )); do c=$((a + b))
echo -n " $c" a=$b b=$c
done echo

```

```

echo "Enter number of terms:" read n fibonacci
$ n (d) Check Prime or Composite #!/bin/bash
is_prime() { num=$1 if [ $num -lt 2 ]; then echo
"$num is neither prime nor composite" return fi
for (( i=2; i*i<=num; i++ )); do if [ $((num % i)) -eq
0 then echo "$num is composite" return
fi done echo
"$num is prime"

```

```

echo "Entera number:" read num
is_prime $num (e) Convert
Decimal to Binary #!/bin/bash
decimal_to_binary() { num=$1
binary=""while [ $num -gt 0 ]; do
binary=$((num % 2))$binary
num=$((num / 2)) done echo
"Binary: $binary"

```

```
echo "Enter a decimal number:"  
read num decimal_to_binary  
$num
```

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